

Application of Transfer Learning in Optimized Filter-Bank Regularized CSP to Classification of EEG Signals with Small Dataset

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Abstract— Brain-Computer Interface (BCI) systems are being used by researchers in the field to connect the brain to physical devices, like Electroencephalography (EEG) signals. They are being extensively analyzed with respect to brain electrical activities. EEG is a low-cost brain cognition and imaging technique with high temporal resolutions that can be used to extract features from Motor Imagery tasks. The common spatial pattern (CSP) and its optimized algorithms are effective methods for discriminating and classifying EEG signals. To classify motor imagery tasks in EEG signals, we must use the CSP algorithm to extract features and discriminate spatial patterns associated with movement activities in two-class motor imagery signals. Furthermore, owing to the amount of noise in EEG signals and the limited number of trials per subject, along with a penalty term in the denominator of the CSP cost function, we may optimize the conventional CSP algorithm. In this study, due to differences in each subject's neural activities, we employed transfer learning which used the information for other subjects to regulate features of the subject. Additionally, BCI Competition III dataset IVa was analyzed. Furthermore, this study presents the optimized Filter Bank Regularized CSP algorithm with Transfer Learning to perform the classification of the electroencephalography (EEG) motor imagery signals. Moreover, to compare the efficiency of the proposed algorithm, the conventional CSP and the proposed optimized CSP have been weighed, and results for both methods are presented. The results at the end explain that the classification with 10-fold cross-validation in comparison with that of the proposed method achieves approximately 15% and 21% higher accuracy against the R-CSP and conventional CSP, respectively.

Keywords— Brain-Computer-Interface, BCI, EEG, Motor Imagery, FBRCSP, Common Spatial Pattern,

I. INTRODUCTION

In the last decade, the brain-computer interface (BCI) has been a hot topic in neuroscience, brain science, electrical, computer, and biomedical engineering. Noninvasive BCI systems have been developed and increased researchers' attention in recent years. The brain-computer interface (BCI) transfers and interprets the brain signals and human intentions into external device commands and controls, which cause new ways of interacting between humans and the external environment[1], [2]. These provide a path for people with neuronal, physical, and neuromuscular disabilities to communicate with the outside

world. Additionally, this technique is helpful in rehabilitation methods and studies like patients with motor disabilities and upper body paralysis [3], [4]. The BCI system is based on decoding brain signals using signal processing, feature extraction, machine learning, and pattern recognition algorithms, converting EEG signals to control commands in controlling the related devices. Among various brain mapping methods like fNIRS[5], fMRI, and EEG, the scalp electroencephalogram (EEG) is affordable with low cost and high time resolution, broadly used in BCI systems. Next, to achieve higher performance, hybridization with fNIRS is another option to achieve higher performance in BCI systems[6]–[8]. Motor imagery is a field of study focused on the state of the central or sensory-motor cortex area of the brain[9]–[11]. In motor imagery, a subject starts to imagine a certain task, leading to an intention to move a particular body member like the hand, finger, or tongue without a physical, external, or even muscle movement. This state of imagination leads to neural activities similar to natural movements but with different patterns based on the type of imagination. It can be beneficial for individuals who have suffered a mental illness to engage in activities like arm movement, word spelling, or cursor control. Motor imagery EEG signals are weak signals due to low signal-to-noise ratio (SNR) in other words high level of noise and artifacts. These artifacts may include such sources as (50 or 60 Hz) power frequency of current electricity, physiological signals which act like artifacts in EEG MI signal analysis such as the electromyography (EMG), electrooculography (EOG), and Electrocardiography (ECG)[11]. In addition, MI applications do not require the subject's stimulation in tasks.

MI studies mainly analyze and classify subjects independently[12]. However, in this study, models are trained for every subject precisely and used other subjects' information to regularize the CSP algorithm to reduce the possibility of overfitting and robustness. BCI systems based on motor imagery were used to detect event-related desynchronization (ERD) and event-related synchronization (ERS) across the motor cortex area. Moreover, the ERD/ERS signals vary per subject and location on the brain and frequency domain[13]. Thus, features extraction in EEG signals must be done

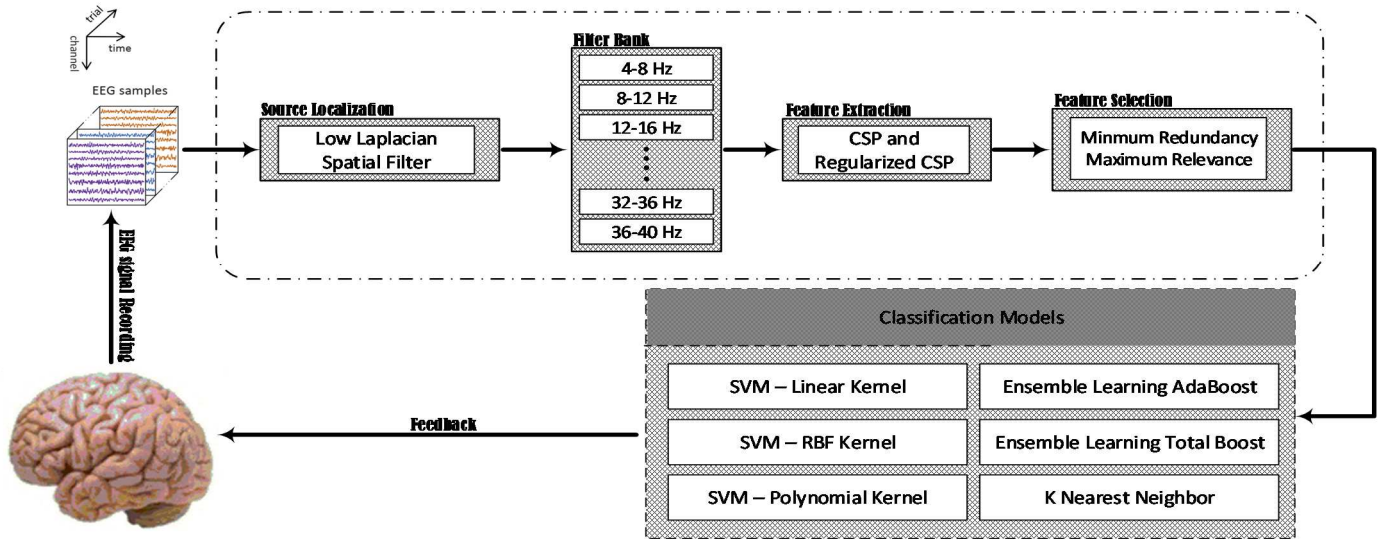


Figure 1. The Proposed Method's Block Diagram

separately for each subject. Based on recently published research, Common Spatial Pattern (CSP) achieved outstanding performance in binary class motor imagery tasks of EEG signals [2],[14]. The spatial filters leading to the possibility of achieving a higher classification accuracy. There exist methods of optimized Common Spatial Patterns leading to more discriminative features in two-class motor imagery EEG signal trials, such as the CCSP[11][15], TRCSP[15] and RCSP[14]. These methods were invented to reduce the disadvantages of conventional CSP, such as the FBCSP[14] proposed to reduce highly dependent frequency bands of spectral patterns.

This paper is structured as follows: Section II discusses the proposed method, Section III discusses the optimized filter bank regularize CSP, Section IV discusses the dataset and experiments used in this study, Section V analyzes the data and compares the proposed method to conventional methods, and Section VI summarizes the findings and makes suggestions for future studies.

II. METHODOLOGY

In this study, the proposed method, as depicted in Fig. 1 is based on the filter bank Regularized CSP algorithm[13]. The proposed method will remove the dependency on the frequency band in the conventional CSP algorithm. However, As EEG signals contain noise and artifacts, the conventional CSP approach may lead in classifications that are overfit. Several processes are necessary to prepare the raw EEG data for further analysis and classification. The proposed approach has five steps as shown in Fig. 1. Finally, we developed classifiers with different kernels to discriminate trials and compare their performance.

1) Source localization

Step 1. In EEG signal dataset with 118 channels, we performed a spatial filter method based on a lower Laplacian spatial filter. The lower Laplacian spatial filter was discussed in[16]. The common average reference(CAR) and large Laplacian spatial

filters yield higher signal-to-noise ratios that cause higher performance in EEG signals classification[16],[17]. The lower Laplacian spatial filter performs like a high-pass filter that enhances localized activity while suppressing diffusion activity. The sum of the weighted potentials of the surrounding electrodes is subtracted from the current electrode potential to generate low Laplacian spatial filters. Inversely proportional to the distance between each electrode and the electrode of interest.

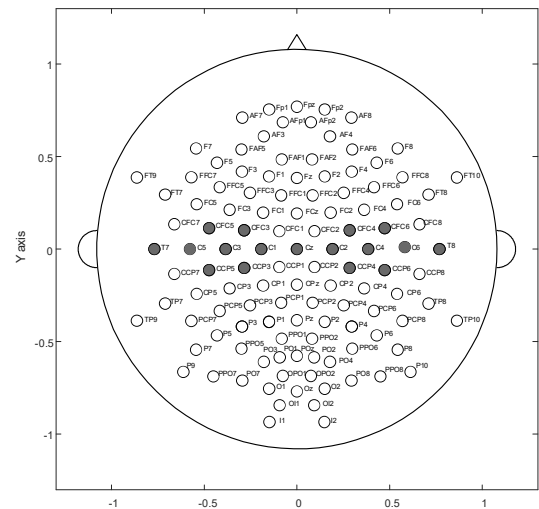


Figure 2. Selected Channels in Motor Cortex Area

2) Channel Selection

Step 2. To classify motor imagery tasks, we just require sensory-motor electrodes inserted on the scalp. Therefore, based on locations in[13], We got 18 electrodes. The electrodes selected are depicted as solid circles in Fig. 2, while the electrodes not selected are shown as white circles.

3) Filter Bank EEG Signal

Step 3. Sensorimotor rhythms (SMR) are oscillations between 8 and 30 Hz in EEG data obtained from the scalp across the sensorimotor cortex area of the brain[18], which change with movement. Frequency domains impact RCSP derived features. To enhance classification performance, each trial should be filtered before using regularized CSP feature extraction. Each EEG channel filtered with 7th-order bandpass Butterworth filter to remove non-related motor imagery and noise. Motor imagery can cause ERS and ERD phenomena: signal power changes in specific frequency bands like μ (8-12 Hz) and β (18-26 Hz) rhythms. Its performance is frequency-dependent and is determined by the EEG's operational frequency band. We employed a filter bank with nine frequency bands in this study. Figure 1 illustrates the nine frequency domains of a filter bank. Additionally, nine Butterworth filters with specified cut-off frequencies were designed. Following that, Regularized CSP features then applied on each of these bands. Then, using a feature selection algorithm, the best features are chosen.

4) EEG Signal Segmentation

Step 4. The proposed CSP algorithm will perform to crop the EEG signals based on the start and end time of trials, which means cutting the signal from the start time to the end time of the signal along with 18 selected channels, referred to as trials. In this study, to crop signals in 18 channels, the start point is set at 0 sec, and the endpoint is set at 3.5 sec based on Fig. 3, which illustrates the structure of a trial.

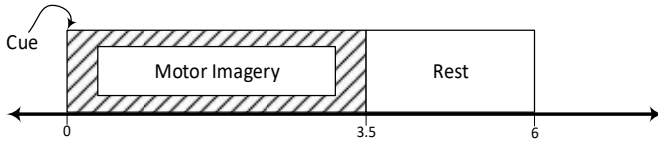


Figure 3. Timing Paradigm of a Single Trial

5) Apply Optimized Regularized CSP method

Step 5. To optimize Regularized CSP method, we need the three parameters α , β , and γ . These three sets of parameters are shown below[14][13].

$$\alpha = \{10^{-10}, 10^{-9}, 10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$$

$$\beta = \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$$

$$\gamma = \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$$

Next, we should find the best optimum parameters to achieve the best optimum performance. Then, to select the best subset features extracted from the Regularized CSP, we select the best features according to the minimum redundancy maximum relevance (mRmR) algorithm[19]. The Regularized CSP consists of $N \times d$ trials (N is the number of features from RCSP and d is the number of filters performed on the signal).

In this method, the best subset of features per specific subject is selected, the results of which give the number of best features selected.

III. OPTIMIZED FILTER BANK REGULARIZED CSP

Table 1. Regularization Parameters

Subject	α	β	γ	Accuracy %
'aa'	10^{-5}	0.1	0.1	95.36
'al'	10^{-7}	0.1	0.2	98.57
'av'	10^{-8}	0	0	85.71
'aw'	10^{-8}	0	0	97.50
'ay'	10^{-7}	0	0	96.43

The goal of the conventional common spatial pattern is to maximize the variance of EEG signal in a class in addition to minimizing the variance of other classes, simultaneously. It means that CSP suits two-class trials. As for the multiclass trials, we could implement the one-vs-one and one-vs-rest methods. Conventional CSP uses spatial filters that extremize the following cost function:

$$J(w) = \frac{w^T C_1 w}{w^T C_2 w} = \frac{w^T X_1 X_1^T w}{w^T X_2 X_2^T w} \quad (1)$$

where T stands for transposition and is the signal trial for class I with rows representing channels and columns representing time points. The cost function in Eq. (1) is the spatial covariance matrix of class i. While the objective is to maximize the cost function, and despite the conventional CSP's popularity and performance, it does have some drawbacks. Among the acknowledged drawbacks is a high sensitivity to noise, which results in overfitting, particularly in datasets with a limited number of trials. As a workaround, we employ the Regularized CSP method as follows:

$$J_{P_{\{1,2\}}}(w) = \frac{w^T \tilde{C}_{\{1,2\}} w}{w^T \tilde{C}_{\{2,1\}} w + \alpha P(w)} \quad (2)$$

The spatial covariance matrix for each of the two classes must be estimated using conventional CSP. However, if the EEG trial set is noisy or the trial count is low, the computed covariance matrix may overfit, resulting in poor discrimination results. As a result, we added a penalty term to the cost function to normalize the findings and reduce the effect of overfitting. The term $\alpha P(w)$ in this new cost function represents the penalty term that is added to the denominator, as follows:

$$P(w) = w^T K w \quad (3)$$

$$\hat{C}_c = (1 - \beta) s_c C_c + \beta G_c \quad (4)$$

$$\tilde{C}_c = (1 - \gamma) \hat{C}_c + \gamma I \quad (5)$$

$$G_c = \sum_{i \in \Omega} \frac{N_c^i}{N_{t,c}} C_c^i \quad (6)$$

$$s_c = \frac{N_c}{N_{t,c}} \quad (7)$$

Table 2. Subjects Performance Comparison

<i>Subject</i>	<i>CSP</i>	<i>Regularized CSP</i>	<i>Optimized FB-R CSP</i>
'aa'	75.89	75.89	95.36
'al'	91.07	91.07	98.57
'av'	56.63	60.2	85.71
'aw'	66.51	70.98	97.50
'ay'	79.36	88	96.43
<i>mean</i>	73.89	77.22	94.714

where C_c is the initial spatial covariance matrix for class c , $\tilde{C}_c$ is the regularized estimate, I is the identity matrix, s_c is a constant scaling parameter. Eq. (6) is a generic covariance matrix to estimate the more robust covariance matrix $\hat{C}_c$ [14]. This generic matrix represents a given prior data based on all subjects. The covariance matrix is based on several subjects previously recorded. This is an effective method to regularize CSP cost function based on subject-to-subject transfer. This method is based on replacing $C1$ and $C2$ used in CSP by their regularized estimates.

$$Z = W^T \cdot x \quad (8)$$

$$f_p = \log \left(\frac{\text{var}(Z)}{\sum \text{var}(Z)} \right) \quad (9)$$

In Eq. (8) we applied the computed weights W on EEG signals x , giving Z as the mapped EEG trials.

Finally, we extract features based on log power features in Eq. (9), where $\text{var}(\hat{z})$ is the variance of $\hat{z}$ over the domain samples [14]. Then, $Z = [Z_1, Z_2, Z_3, \dots, Z_n]^T$, where we choose the n -th first and last $\hat{z}_n$ as the feature set. Classification is then performed on the selected features.

IV. DATA DESCRIPTION

In this study, the selected EEG dataset to be studied consisted of a small number of data is the BCI Competition III Dataset IVa [13]. This publicly available dataset contains data from five healthy volunteers (identified as 'aa', 'al', 'av', 'aw', and 'ay') who were requested to execute cued motor imagery-based BCI experiments in three classes, including right hand, left hand, and right foot motion. Additionally, this dataset contains just right hand and right foot trials. The EEG signals were acquired using a BrainAmp amplifier and a 128-channel Ag/AgCl electrode cap, albeit only 118 channels were available in this dataset. The EEG signals were digitized and amplified at 1000Hz, and then resampled to 100Hz after filtering. Additionally, the cap's electrode placement is based on the international 10/20 method. Each trial began with a 3.5-second visual cue indicating the MI task. After 1.75 to 2.25 seconds of rest and relaxation, the subject's EEG was recorded from 118 channels or electrodes. Each subject had access to a total of 280 trials.

V. RESULTS AND DISCUSSION

We employed two primary ways to validate the proposed algorithm: first, we compared its findings to those produced by other researchers. Second, we used k -fold cross validation to

verify our findings. It is based on partitioning a dataset into training and test data when the number of trials is limited. Following that, this method is repeated to ensure that the results are accurate. The accuracy of classification in the BCI Competition III dataset IVa using the Transfer Learning in Filter Bank Regularized CSP algorithm is shown in Table 2. This table also shows a comparison of results from other methods including conventional CSP proposed in paper [14] and Regularized CSP proposed in paper [13]. When compared to the results of the regularized CSP technique, the proposed method improves classification accuracy by around 15%. On the basis of these findings, we infer that the Transfer learning concept produces superior results. It also enhances performance by roughly 21% when compared to conventional CSP. Remain aware that the results in Table 2 are based on a support vector machine with radial basis function that validates using a 10 fold cross-validation procedure. Additionally, the results presented here are based on improved coefficients as mentioned in Table 1. The proposed method uses Transfer Learning to compute a generic matrix as described in the earlier section. The dependency on the computed generic matrix increases when we have a limited number of data, and consequently, the performance and classification accuracy vary with parameters, hence requiring these parameters to be tuned. Based on the results described in table 2, the proposed method yields higher accuracy in all five subjects, and except for subject 'av,' the rest yield an accuracy above 90%, which is a considerable achievement in BCI systems. Additionally, according to table 1, we should find three coefficients based on the described proposed method. The results demonstrate that the β and γ parameters are very close to zero to yield optimum results. The α alpha parameter is a regularization term in the CSP algorithm. The results show that the regularization term causes better results in feature extraction EEG signals. Based on table 3, classification methods applied on EEG signals consist of support vector machine (SVM) includes linear, radial basis function, and polynomial kernel functions, in addition to k nearest neighbor, with k equal to 5 to discriminate features to five neighbors based on specified density function. Moreover, implementation of ensemble learning methods leads to higher accuracy in classification with the help of the combination of multiple classifiers with Adaboost ensemble learning method, as well as the total boost ensemble learning method, each of which consists of decision tree as the kernel function. We implemented these two classifiers with $m=200$, which means a combination of 200 classifiers to obtain higher and more robust classification accuracy on the dataset. The results given in table 3 show the accuracy of each classifier for every subject. These

Table 4. Classifiers Performance Comparison

Classification model	'aa'	'al'	'av'	'aw'	'ay'	Average
Linear SVM	92.14	97.86	86.43	94.64	93.57	92.93
Polynomial SVM	95	98.57	84.29	97.14	95	94
RBF SVM	95.36	98.93	85.71	96.79	96.43	94.64
K Nearest Neighbor	91.79	97.14	82.5	95.36	93.93	92.14
Adaboost Tree	94.29	94.64	85	95.36	94.64	92.79
Total Boost Tree	95.36	93.93	84.29	96.43	95	93
Average	93.99	96.85	84.7	95.95	94.76	-

results illustrate that both total boost and AdaBoost ensemble methods yield high accuracies. As for the 'aa', 'aw', and 'ay' subjects, the total boost ensemble method gives a higher accuracy in comparison with that of the AdaBoost ensemble method. Furthermore, in 'al' and 'av' subjects, AdaBoost ensemble methods yield a higher accuracy. The results given in table 3 are obtained by the support vector machine classification methods with 'Linear', 'Polynomial', and 'RBF' kernel functions. For comparison purposes, it is noted that the accuracies obtained for subjects 'aa', 'al', and 'ay' are higher in SVM with the RBF kernels, while for the 'av' subject, the linear kernel results in a higher accuracy, and for the 'aw' subject, the polynomial SVM yields a higher classification accuracy.

VI. CONCLUSION

The suggested method overcomes the limitations of the current CSP method, such as sample size and frequency band dependence. To tackle these challenges, we employed filter banks with nine frequency bands and retrieved two features from each. Due to the dependency on frequency bands, using filter banks in CSP improves results. We also built the Regularization approach, which adds a penalty term to the CSP function's denominator and computes a generic matrix from all subjects. The purpose of this study is to enhance performance degradation owing to overfitting in the restricted number of trials. According to the data, employing this proposed technique enhances classification accuracy by around 21% compared to using the conventional CSP with 70% accuracy. There is a notable lack of study in the literature on tackling other challenges of the CSP approach, classifying motor imagery EEG signals, and studies dealing with future problems in BCI systems. As a result, more research is needed to improve accuracy and results.

VII. REFERENCES

- [1] L. F. Nicolas-Alonso and J. Gomez-Gil, "Brain computer interfaces, a review," *Sensors*, vol. 12, no. 2, pp. 1211–1279, 2012, doi: 10.3390/s120201211.
- [2] F. Lotte *et al.*, "A review of classification algorithms for EEG-based brain-computer interfaces: A 10 year update," *J. Neural Eng.*, vol. 15, no. 3, 2018, doi: 10.1088/1741-2552/aab2f2.
- [3] F. Yger, M. Berar, and F. Lotte, "Riemannian Approaches in Brain-Computer Interfaces: A Review," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 10, pp. 1753–1762, 2017, doi: 10.1109/TNSRE.2016.2627016.
- [4] H. Raza, A. Chowdhury, and S. Bhattacharyya, "Deep Learning based Prediction of EEG Motor Imagery of Stroke Patients' for Neuro-Rehabilitation Application," *Proc. Int. Jt. Conf. Neural Networks*, no. Mi, 2020, doi: 10.1109/IJCNN48605.2020.9206884.
- [5] N. Naseer and K.-S. Hong, "fNIRS-based brain-computer interfaces: a review," *Front. Hum. Neurosci.*, vol. 9, Jan. 2015, doi: 10.3389/fnhum.2015.00003.
- [6] P. A. Cicalese *et al.*, "An EEG-fNIRS hybridization technique in the four-class classification of alzheimer's disease," *J. Neurosci. Methods*, vol. 336, no. February, p. 108618, 2020, doi: 10.1016/j.jneumeth.2020.108618.
- [7] S. Mandal, B. K. Singh, and K. Thakur, "Classification of working memory loads using hybrid EEG and fNIRS in machine learning paradigm," *Electron. Lett.*, vol. 56, no. 25, pp. 1386–1389, 2020, doi: 10.1049/el.2020.2710.
- [8] P. A. Cicalese *et al.*, "An EEG-fNIRS hybridization technique in the four-class classification of alzheimer's disease," *J. Neurosci. Methods*, vol. 336, no. November 2019, p. 108618, 2020, doi: 10.1016/j.jneumeth.2020.108618.
- [9] M. Ahn and S. C. Jun, "Performance variation in motor imagery brain-computer interface: A brief review," *J. Neurosci. Methods*, vol. 243, pp. 103–110, 2015, doi: 10.1016/j.jneumeth.2015.01.033.
- [10] N. Padfield, J. Zabalza, H. Zhao, V. Masero, and J. Ren, "EEG-based brain-computer interfaces using motor-imagery: Techniques and challenges," *Sensors (Switzerland)*, vol. 19, no. 6, pp. 1–34, 2019, doi: 10.3390/s19061423.
- [11] S. Aggarwal and N. Chugh, "Signal processing techniques for motor imagery brain computer interface: A review," *Array*, vol. 1–2, no. August, p. 100003, 2019, doi: 10.1016/j.array.2019.100003.
- [12] N. Samiee, S. H. Sardouie, H. Mohammad, and Foroughmand-Aarabi, "A Transfer Learning Algorithm Based on CSP Regularizations of Recorded EEG for Between-Subject Classification," *2019 26th Natl. 4th Int. Iran. Conf. Biomed. Eng. ICBME 2019*, no. November, pp. 199–203, 2019, doi: 10.1109/ICBME49163.2019.9030388.
- [13] S. H. Park, D. Lee, and S. G. Lee, "Filter Bank Regularized Common Spatial Pattern Ensemble for Small Sample Motor Imagery Classification," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 26, no. 2, pp. 498–505, 2018, doi: 10.1109/TNSRE.2017.2757519.
- [14] F. Lotte and C. Guan, "Regularizing common spatial patterns to improve BCI designs: Unified theory and new algorithms," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 2, pp. 355–362, 2011, doi: 10.1109/TBME.2010.2082539.
- [15] K. D. Ghanbar, T. Y. Rezaii, A. Farzamnia, and I. Saad, "Correlation-based common spatial pattern (CCSP): A novel extension of CSP for classification of motor imagery signal," *PLoS One*, vol. 16, no. 3 March 2021, pp. 1–18, 2021, doi: 10.1371/journal.pone.0248511.
- [16] S. Tsuchimoto *et al.*, "Use of common average reference and large-Laplacian spatial-filters enhances EEG signal-to-noise ratios in intrinsic sensorimotor activity," *J. Neurosci. Methods*, vol. 353, no. December 2020, p. 109089, 2021, doi: 10.1016/j.jneumeth.2021.109089.
- [17] D. J. McFarland, L. M. McCane, S. V. David, and J. R. Wolpaw, "Spatial filter selection for EEG-based communication," *Electroencephalogr. Clin. Neurophysiol.*, vol. 103, no. 3, pp. 386–394, 1997, doi: 10.1016/S0013-4694(97)00022-2.
- [18] J. Lu, D. J. McFarland, and J. R. Wolpaw, "Adaptive Laplacian filtering for sensorimotor rhythm-based brain-computer interfaces," *J. Neural Eng.*, vol. 10, no. 1, 2013, doi: 10.1088/1741-2560/10/1/016002.
- [19] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information: Criteria of Max-Dependency, Max-Relevance, and Min-Redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226–1238, 2005, doi: 10.1109/TPAMI.2005.159.